Retail Analytics

Formula & KPI Reference Guide

Comprehensive formulas, thresholds, and classification rules
for retail data analysis and performance reporting

Document Version	3.1
Applicable Data	Weekly Sales, Inventory, Store Performance
Units	All monetary values in USD unless stated
Time Basis	Weekly / Monthly / Annual as specified per formula
Last Revised	Q1 2025

Table of Contents

1. Sales Performance Metrics

Sales performance metrics quantify how effectively a retail store or chain converts inventory and customer traffic into revenue. These formulas apply to the weekly sales data available in the Sales DataFrame.

1.1 Total Weekly Sales (Store Level)

Total_Sales = SUM(Weekly_Sales) for each Store

Weekly_Sales : numeric — sales revenue for that week

Store : integer — store identifier

Note: Aggregate over all date rows per Store to get lifetime or period total.

1.2 Average Weekly Sales

Avg_Weekly_Sales = Total_Sales / Number_of_Weeks

Total_Sales : numeric — sum of all weekly sales for the store/period

Number_of_Weeks: integer — count of distinct weeks in the period

1.3 Sales Per Square Foot

Sales_PSF = Total_Annual_Sales / Store_Area_SqFt

Total_Annual_Sales : numeric — annual revenue

Store_Area_SqFt : numeric — gross leasable area in square feet

Note: Benchmark: Top quartile stores exceed $350/sqft annually.

1.4 Week-over-Week (WoW) Sales Growth

WoW_Growth (%) = ((Sales_Week_N - Sales_Week_N-1) / Sales_Week_N-1) x 100

Sales_Week_N : numeric — current week sales

Sales_Week_N-1 : numeric — prior week sales

Note: Flag any WoW decline greater than 15% for store manager review.

1.5 Month-over-Month (MoM) Sales Growth

MoM_Growth (%) = ((Sales_Month_N - Sales_Month_N-1) / Sales_Month_N-1) x 100

Sales_Month_N : numeric — current month total sales

Sales_Month_N-1 : numeric — prior month total sales

2. Growth & Trend Analysis

2.1 Compound Annual Growth Rate (CAGR)

CAGR (%) = ((End_Value / Start_Value) ^ (1 / Number_of_Years) - 1) x 100

End_Value : numeric — final period sales value

Start_Value : numeric — base period sales value

Number_of_Years: numeric — number of years between Start and End

Note: Use fiscal year totals. Number_of_Years = (End_Year - Start_Year). CAGR above 8% is considered strong growth for mature retail.

2.2 Year-over-Year (YoY) Growth

YoY_Growth (%) = ((Sales_Year_N - Sales_Year_N-1) / Sales_Year_N-1) x 100

Sales_Year_N : numeric — current year total sales

Sales_Year_N-1 : numeric — prior year total sales

2.3 Rolling 4-Week Moving Average

MA4 = (Sales_W + Sales_W-1 + Sales_W-2 + Sales_W-3) / 4

Sales_W : numeric — sales in week W (current)

Sales_W-k : numeric — sales k weeks prior

Note: Use rolling(4).mean() in pandas. Smooths holiday and promotional spikes.

2.4 Sales Trend Index (STI)

STI = (Current_Period_Sales / Baseline_Period_Sales) x 100

Current_Period_Sales : numeric — sales in evaluation period

Baseline_Period_Sales : numeric — sales in reference baseline period

Note: STI = 100 means flat. STI > 110 = growth. STI < 90 = decline. Baseline is typically same period prior year.

3. Inventory & Supply Chain Metrics

3.1 Inventory Turnover Ratio

Inventory_Turnover = Cost_of_Goods_Sold / Average_Inventory

Cost_of_Goods_Sold : numeric — COGS for the period

Average_Inventory : numeric — (Opening_Inventory + Closing_Inventory) / 2

Note: Retail benchmark: 4–6 turns per year. Below 4 indicates overstocking.

3.2 Days Sales of Inventory (DSI)

DSI = (Average_Inventory / Cost_of_Goods_Sold) x 365

Average_Inventory : numeric — mean inventory value over period

Cost_of_Goods_Sold : numeric — annual COGS

Note: Target DSI < 60 days. DSI > 90 days triggers markdown review.

3.3 Stock-to-Sales Ratio

Stock_to_Sales = Inventory_Value / Net_Sales

Inventory_Value : numeric — current inventory at retail price

Net_Sales : numeric — net sales for the same period

Note: Healthy range: 2.0 to 4.0. Above 5.0 signals excess inventory risk.

4. Profitability & Margin Formulas

4.1 Gross Profit Margin

Gross_Margin (%) = ((Net_Sales - COGS) / Net_Sales) x 100

Net_Sales : numeric — revenue after returns

COGS : numeric — cost of goods sold

Note: Retail average: 25–45%. Grocery: 20–25%. Apparel: 40–60%.

4.2 Net Profit Margin

Net_Margin (%) = (Net_Income / Net_Sales) x 100

Net_Income : numeric — revenue minus all expenses and taxes

Net_Sales : numeric — total revenue after returns

4.3 Return on Sales (ROS)

ROS (%) = (Operating_Profit / Net_Sales) x 100

Operating_Profit : numeric — EBIT (earnings before interest and tax)

Net_Sales : numeric — net revenue

Note: ROS below 3% triggers cost structure review.

4.4 Break-Even Sales Volume

Break_Even_Units = Fixed_Costs / (Selling_Price - Variable_Cost_per_Unit)

Fixed_Costs : numeric — rent, salaries, utilities

Selling_Price : numeric — average unit selling price

Variable_Cost_per_Unit : numeric — COGS per unit

5. Customer & Basket Metrics

5.1 Average Transaction Value (ATV)

ATV = Total_Sales / Number_of_Transactions

Total_Sales : numeric — gross sales in period

Number_of_Transactions: integer — count of distinct sales transactions

Note: Benchmark varies by format. Hypermarket ATV: $80–$150. Convenience: $8–$20.

5.2 Average Units Per Transaction (UPT)

UPT = Total_Units_Sold / Number_of_Transactions

Total_Units_Sold : integer — total items sold

Number_of_Transactions: integer — total transaction count

5.3 Customer Conversion Rate

Conversion_Rate (%) = (Number_of_Transactions / Foot_Traffic) x 100

Number_of_Transactions: integer — paying customers

Foot_Traffic : integer — total store visitors

Note: Target: above 30%. Below 20% triggers merchandising and layout review.

5.4 Customer Lifetime Value (CLV)

CLV = Avg_Purchase_Value x Purchase_Frequency x Customer_Lifespan_Years

Avg_Purchase_Value : numeric — mean spend per visit

Purchase_Frequency : numeric — visits per year

Customer_Lifespan_Years: numeric — years customer remains active

6. Risk Thresholds & Classification Rules

The following thresholds are used to classify store performance and flag records in the sales dataset for operational review. Apply these rules to Weekly_Sales and derived metrics columns.

6.1 Weekly Sales Risk Classification

Category	Range / Condition	Action / Flag
Critical Low	Weekly_Sales < 10,000	Immediate escalation to regional manager
Low	10,000 – 24,999	Flag for store manager review
Normal	25,000 – 99,999	No action required
High	100,000 – 199,999	Positive flag — monitor for consistency
Exceptional	>= 200,000	Audit for data accuracy; log as outlier

6.2 WoW Growth Rate Flags

Category	Range / Condition	Action / Flag
Severe Decline	WoW_Growth < -20%	Alert: investigate cause immediately
Moderate Decline	-20% to -10%	Warning: review promotions and stock
Stable	-10% to +10%	Normal operating range
Strong Growth	+10% to +30%	Positive: document contributing factors
Spike	> +30%	Verify data; likely promotional event

6.3 Inventory Turnover Classification

Category	Range / Condition	Action / Flag
Excess Stock	Turnover < 3.0	Initiate markdown / clearance
Slow Moving	3.0 – 3.9	Review replenishment orders
Optimal	4.0 – 6.0	Target range — maintain
Lean	6.1 – 8.0	Monitor for stockout risk
Critically Lean	> 8.0	Urgent reorder — stockout imminent

6.4 Gross Margin Classification

Category	Range / Condition	Action / Flag
Below Threshold	Gross_Margin < 15%	Cost structure review required

Acceptable	15% – 24%	Minimum acceptable — monitor
Target	25% – 44%	Healthy retail margin
High	45% – 59%	Premium / specialty category
Exceptional	>= 60%	Validate COGS data accuracy

7. Seasonality Adjustment Methodology

Retail sales exhibit strong seasonal patterns (holiday peaks, back-to-school, summer slowdowns). Seasonally adjusted figures allow fair comparison across different time periods.

7.1 Seasonal Index Calculation

Seasonal_Index (week W) = Avg_Sales_in_Week_W_across_Years / Overall_Weekly_Avg

Avg_Sales_in_Week_W : numeric — mean sales in calendar week W over all years

Overall_Weekly_Avg : numeric — grand mean of all weeks across all years

Note: Calculate one index per calendar week (1–52). Index > 1.0 means above-average week.

7.2 Seasonally Adjusted Sales

Adjusted_Sales = Actual_Sales / Seasonal_Index

Actual_Sales : numeric — raw Weekly_Sales value

Seasonal_Index : numeric — index for that calendar week (from 7.1)

Note: Use Adjusted_Sales for YoY or WoW comparisons across different seasonal periods.

7.3 Holiday Week Multiplier

Holiday_Multiplier = Holiday_Week_Sales / Same_Week_Non_Holiday_Avg

Holiday_Week_Sales : numeric — sales during the holiday week

Same_Week_Non_Holiday_Avg: numeric — average sales in that calendar week in non-holiday years

Note: Weeks 51 and 52 (Christmas) typically show multipliers of 1.8 to 2.5. Week 47 (Thanksgiving in USA) multiplier: 1.4 to 1.9.

7.4 Known Seasonal Adjustment Factors (Reference)

Period	Seasonal Index	Reason
Week 1–4 (Jan)	0.78	Post-holiday slowdown
Week 5–12 (Feb–Mar)	0.85	Winter baseline
Week 13–20 (Apr–May)	0.95	Spring uptick
Week 21–30 (Jun–Jul)	1.05	Summer peak (AC, outdoor)
Week 31–38 (Aug–Sep)	1.00	Back-to-school
Week 39–46 (Oct–Nov)	1.10	Pre-holiday build
Week 47 (Thanksgiving)	1.55	Thanksgiving week
Week 48–50 (Dec early)	1.35	Holiday shopping
Week 51–52 (Christmas)	1.90	Peak holiday week

8. Store Performance Scoring

The Store Performance Score (SPS) is a composite index (0–100) combining sales, growth, and efficiency metrics. It is used for quarterly store rankings and resource allocation decisions.

8.1 Store Performance Score (SPS)

SPS = (0.40 x Sales_Score) + (0.30 x Growth_Score) + (0.20 x Margin_Score) + (0.10 x Efficiency_Score)

Sales_Score : 0–100 — normalized total sales rank within peer group
Growth_Score : 0–100 — based on YoY growth rate (see 8.2)
Margin_Score : 0–100 — based on gross margin percentile
Efficiency_Score : 0–100 — based on Sales_PSF rank

Note: Weights: Sales 40%, Growth 30%, Margin 20%, Efficiency 10%.

8.2 Growth Score Mapping

Growth_Score = MIN(100, MAX(0, 50 + (YoY_Growth x 2.5)))

YoY_Growth : numeric (%) — year-over-year growth rate from formula 2.2

Note: Growth of 0% → score 50. Growth of +20% → score 100. Decline of -20% → score 0.

8.3 SPS Performance Bands

Band	Score Range	Management Action
Elite	SPS >= 85	Top performer — expand resource allocation
Strong	70 – 84	Above average — maintain strategy
Average	50 – 69	Meets baseline — incremental improvement plan
At Risk	35 – 49	Below average — mandatory action plan required
Critical	< 35	Performance intervention — possible closure review

9. Anomaly Detection Rules

The following statistical rules define when a data point should be flagged as an anomaly in the weekly sales dataset. Anomalies may indicate data entry errors, fraud, system issues, or genuine exceptional events.

9.1 Z-Score Anomaly Detection

Z_Score = (Weekly_Sales - Mean_Sales) / Std_Dev_Sales

Weekly_Sales : numeric — the observation being tested

Mean_Sales : numeric — mean of Weekly_Sales over the evaluation window

Std_Dev_Sales : numeric — standard deviation of Weekly_Sales

Note: Flag if |Z_Score| > 3.0. Use a rolling 52-week window for Mean and Std_Dev.

9.2 Z-Score Threshold Classification

Category	Condition	Action
Normal	|Z| <= 2.0	No flag — expected variation
Watch	2.0 < |Z| <= 2.5	Soft flag — monitor next week
Investigate	2.5 < |Z| <= 3.0	Medium flag — analyst review
Anomaly	|Z| > 3.0	Hard flag — requires explanation or correction

9.3 IQR-Based Outlier Rule

Lower_Bound = Q1 - 1.5 x IQR Upper_Bound = Q3 + 1.5 x IQR IQR = Q3 - Q1

Q1 : numeric — 25th percentile of Weekly_Sales distribution

Q3 : numeric — 75th percentile of Weekly_Sales distribution

IQR : numeric — interquartile range (Q3 minus Q1)

Note: Flag any Weekly_Sales below Lower_Bound or above Upper_Bound as outlier.

9.4 Holiday Flag Rule

Is_Holiday_Week = 1 if Holiday == True else 0

Holiday : boolean — from the IsHoliday column in the dataset

Note: Anomaly detection thresholds are relaxed by 1.5x multiplier for holiday-flagged weeks. Adjust Z-score threshold to 4.5 and IQR multiplier to 2.25 for holiday weeks.

10. Compliance & Audit Thresholds

These thresholds are mandated by internal audit policy. Any record in the dataset that violates these conditions must be flagged in the compliance report and reviewed within 5 business days.

10.1 Mandatory Compliance Flags

Flag Type	Condition	Required Action
Zero Sales Week	Weekly_Sales == 0	Investigate: store closure or data error
Negative Sales	Weekly_Sales < 0	Data error — must correct before reporting
Implausible Spike	Weekly_Sales > 500,000	Manual verification required
Missing Holiday Flag	Date in holiday list but IsHoliday == False	Data integrity issue
Duplicate Week-Store	Same Store + Date appears >1 row	Deduplication required

10.2 Fuel Price Impact Thresholds

Impact Level	Fuel_Price ($/gal)	Sales Adjustment Policy
Low Impact	Fuel_Price < 2.50	Normal operations — no adjustment
Medium Impact	2.50 – 3.49	Monitor discretionary category sales
High Impact	3.50 – 4.49	Apply -5% sales forecast adjustment
Critical Impact	Fuel_Price >= 4.50	Apply -12% adjustment; escalate to category managers

10.3 Temperature Impact on Sales (Outdoor/Seasonal Categories)

Condition	Temperature Range	Merchandising Action
Cold	Temperature < 32 F	Boost winter goods weight by +15%
Cool	32 – 59 F	Standard seasonal mix
Moderate	60 – 79 F	Baseline — no adjustment
Warm	80 – 94 F	Boost cooling and beverage by +10%
Heat Event	Temperature >= 95 F	Boost AC/fan sales flag; flag for inventory alert

10.4 CPI Impact on Sales Planning

Condition	CPI Index Value	Planning Action
Deflation	CPI < 100	Review pricing strategy downward
Stable	100 – 109	Baseline planning — no CPI adjustment

Mild Inflation	110 – 124	Apply +3% to cost forecasts
High Inflation	125 – 149	Apply +8% to cost forecasts; review margins
Severe	CPI >= 150	Immediate pricing committee review

Document End. *For questions regarding formula application or threshold updates, contact the Retail Analytics Centre of Excellence. All formulas are reviewed quarterly. Version history maintained in the analytics wiki.*